

Ye SU

Robotics Application & Mechatronics Engineer

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PROFILE

M.Sc. student in Medical Technology at TUM with a completed M.Sc. in Mechatronics and Robotics and an industrial mechanical engineering background. Hands-on experience in mechanical design, CAD, FDM prototyping, sensor integration, robotic platforms, and experimental system development. Currently working with embodied robots at MINTH-AGIBOT Robotic and developing an instrumented ultrasound system for robot learning at TUM-CAMP(IFL). Interested in robotics application engineering and system integration, particularly hands-on roles combining mechanical design, sensing, robot programming, prototyping, and real-world deployment.

PROFESSIONAL EXPERIENCE

Robotics & After-Sales Intern

05.2026–Present

MINTH Group – European Robotics Team

Munich, Germany

- Operate embodied robotic platforms, including AgiBot A2, G2, X2, and D1, for demonstrations and small-scale application scenarios; support customer visits and technical product demonstrations with the sales team.
- Support robot commissioning and deployment, including system setup, functional checks, customer-specific configuration, operational guidance, user training, and technical handover.
- Perform first-line troubleshooting of robot hardware and software; inspect system logs and device status, conduct initial fault diagnosis, and coordinate complex technical issues with the supplier's engineering teams.
- Manage and track robots, accessories, and demo equipment across the Munich showroom and warehouse locations; maintain asset and equipment-status records and coordinate technical and logistical requirements across operations, logistics, and intelligent solutions teams.

Assistant Engineer – Mechanical Design & Project Support

09.2014–06.2018

North Heavy Industries Group Co., Ltd.

Shenyang, China

- Supported mechanical design of industrial equipment through 3D CAD modelling, interference checks, and preparation of detailed design drawings.
- Supported an international sintering-machine export project, coordinating steel-structure design and technical requirements across engineering, manufacturing, quality, and project teams.
- Worked with international third-party inspectors during on-site manufacturing supervision and supported resolution of technical and quality issues.

SELECTED PROJECTS

Instrumented Freehand Ultrasound System for Robot Learning

04.2026–Present

Chair of Computer Aided Medical Procedures (CAMP), TUM

Munich, Germany

- Designed and iteratively developed an instrumented ultrasound data acquisition system, integrating an ultrasound probe, Intel RealSense RGB-D camera, NDI optical tracking, and a 6-axis force/torque sensor within a custom 3D-printed mechanical assembly.
- Designed and prototyped the mechanical hardware for repeatable assembly, ergonomic freehand operation, sensor integration, and subsequent mounting of the same instrumented device to a robot end-effector.
- Built and validated a multimodal acquisition pipeline using ROS/rosbag to synchronously record ultrasound images, RGB-D data, 6D force/torque measurements, and tracked 6D probe poses during human freehand scanning.
- Develop calibration and data-processing methods, including extrinsic calibration, 3D coordinate transformations, temporal alignment, and pose-dependent gravity compensation for force/torque measurements.
- Process the collected human demonstration data into a structured dataset for subsequent robot-learning experiments and evaluation of robotic reproduction of ultrasound scanning.

One-Shot FDM-Printed Adaptive Robotic Gripper

Institute of Micro Technology and Medical Device Technology (MIMED), TUM

Munich, Germany

- Designed and developed an adaptive robotic gripper with one-shot FDM-printed rigid-flexible hybrid fingers, combining compliant mechanisms with manufacturable mechanical design.

- Developed the gripper mechanism in CAD and fabricated and assembled functional prototypes using FDM 3D printing; iteratively optimized the design through mechanical and grasping experiments.
- Performed kinematic modeling and analysis and applied numerical optimization to determine key structural parameters of the compliant finger mechanism.
- Designed and built a dedicated experimental setup to evaluate mechanical loading and grasping performance; integrated the final prototype with a robotic system for experimental validation and real-world grasping demonstrations.

Photoacoustic NO₂ Sensor – Machine Learning & Experimental Evaluation

Institute for Measurement Systems and Sensor Technology (MST), TUM

Munich, Germany

- Evaluated an existing photoacoustic spectroscopy system for NO₂ measurement and designed the fixture and heat-transfer assembly for the experimental setup.
- Assembled and tested the experimental hardware and performed data acquisition, sensor data processing, and iterative evaluation of the measurement system.
- Evaluated and compared multiple machine-learning models for mapping sensor signals to NO₂ concentrations and analyzed their prediction performance.

EDUCATION

M.Sc. Medical Technology

04.2024–Present

Technical University of Munich

Germany

M.Sc. Mechatronics and Robotics

04.2022–04.2025

Technical University of Munich

Germany

B.Sc. Mechanical Engineering and Automation

09.2010–06.2014

Northeastern University

Shenyang, China

TECHNICAL SKILLS

- **Mechanical Design and Prototyping:** SolidWorks, mechanical design, fixture and component design, technical drawings, FDM 3D printing, rapid prototyping, mechanical assembly, compliant mechanisms
- **Robotics and System Integration:** robot operation and deployment, multimodal sensor integration, 3D coordinate transformations, 6D force/torque sensing, NDI optical tracking, Intel RealSense RGB-D sensing, rosbag, robot troubleshooting
- **Programming and Data:** Python, robotics data processing, Linux, LeRobot dataset preparation, basic PLC and industrial automation knowledge
- **Simulation and Analysis:** Abaqus, SolidWorks Simulation

PUBLICATION

Su, Y.* (2026). **One-Shot FDM-Printed Adaptive Robotic Gripper with Rigid-Flexible Hybrid Fingers.** *Mechatronics (Elsevier)*. DOI: [10.1016/j.mechatronics.2026.103482](https://doi.org/10.1016/j.mechatronics.2026.103482)

LANGUAGES

- **Chinese:** Native
- **German:** Good working knowledge
- **English:** Good working knowledge

SELECTED AWARDS

- Second Prize – National College Student Mechanical Design Competition
- First Prize – Liaoning Mechanical Innovation Design Competition